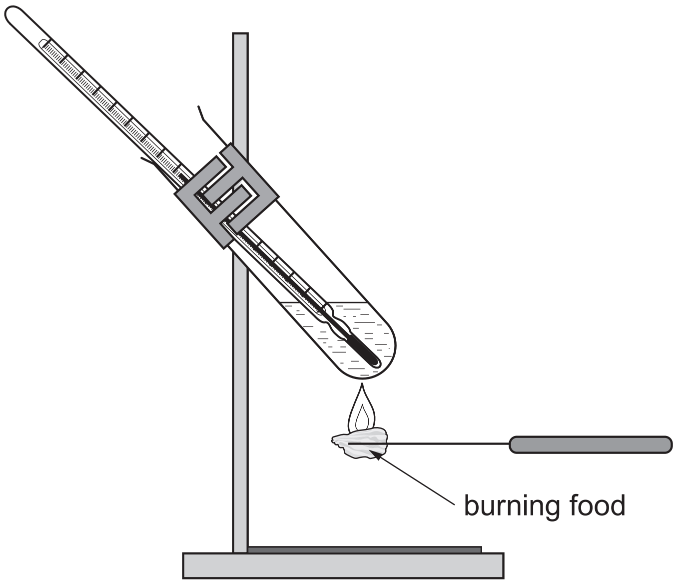
4. *(a)* The apparatus shown in the diagram below can be used to compare the energy content of different foods.

The energy content of the food is released when it is burned and this causes the water to increase in temperature.



You are asked to use the apparatus to compare the energy content of different pasta types of known mass.

(i) The table below shows part of the risk assessment for this investigation.

Complete the table. [2]

Hazard	Risk	Control measure
Apparatus is hot	....	....

(ii) For this investigation, state:

I. the apparatus you would use to find the mass of the pasta; [1]

....

Examiner only

II. **two** factors which you would keep constant throughout the investigation to ensure fair testing; [2]

....

....

III. how you would work out the increase in the temperature of the water. [2]

....

....

....

....

(iii) Explain why it is important to burn each piece of pasta completely. [1]

....

....

(b) The table shows some results from an investigation.

Pasta type	Total mass of pasta burned (g)	Total energy released from pasta (J)	Energy released in Joules per gram (J/g)
plain	9	293.4	32.6
wholewheat	10	282.0	28.2
green	9	252.0	....

(i) Calculate the energy released in Joules per gram for green pasta. **Write your answer in the table.** [2]

Space for working.

(ii) From these results, state which type of pasta has the highest energy content. Give the reason for your answer. [1]

Type of pasta ....

Reason ....

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